

iREPAIR



Company Name			
Block Title		Block Diagram	
Project	Date	Rev	1.0
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I2C Configuration Table

I2C	GPIO	Function	Material	I2C Slave Address(7-bit mode)
SPK_I2C_SDA/CLK	GPIO 4/GPIO 5	Smart PA	FS1862(1st)	0x34
			TAS2558(2nd)	0x4C
		Sar sensor	96T346HW (1st)	0x20
			SX9325(2nd)	0x28
GSENSOR_I3C_SDA/SCL	GPIO 22/GPIO 23	TOF	VL53L1CBV0FY/1	0x52
			LSM6DSOW (1st)	0x6A
		A+G	ICM-42605 (2nd)	0x10
SENSOR_I2C_SDA/SC	GPIO 28/GPIO 29	M sensor	AK09918(1st)	0x0C
			MNC5603NJ(2nd)	0x30
		PS	LTR-2568ALS-WA(1st)	0x23
			TBD(2nd)	TBD
		ALS	TSL2540(1st)	0x39
			MN29005(2st)	0x49

SPI Configuration Table

SPI	Function
QUD1, SE3	LCD interface
QUD1, SE1	FPS interface

I2S Configuration Table

I2S	Function
I2S	AUDIO PA interface



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04.	GPIO MAP	27.	LCD_TP
05.	SM6125 Control/UFS/SDC1/SDC2	28.	REAR_CAM/FLASH
06.	SM6125 LPDDR4	29.	FRONT_CAM/CAM_POWER
07.	SM6125 CSI/DSI	30.	SIM_SD_KEY_LINK_BATT
08.	SM6125 GPIO (Internal Codec)	31.	DTV
09.	SM6125 Power (Core)	32.	NFC
10.	SM6125 Power (Other, LP4X)	33.	PRIMARY_ANTENNA
11.	SM6125 Ground	34.	PA & PA_THERMAL
12.	PM6125 CLKs/Control_GND	35.	TRX_LB
13.	PM6125 Buck Converter1	36.	TRX_MHB
14.	PM6125 Buck Converter2	37.	DIVERSITY_ANT
15.	PM6125 LDO	38.	DRX_MB_HB
16.	PM8008 and Bob	39.	DRX_LB
17.	SIM_SD	40.	SDR660_MSS
18.	PMI632 Control/Display/GPIO	41.	SDR660_PWR
19.	PM660L Display	42.	SDR660_DA/PRX/DRX/FBRX
20.	eMCP	43.	QFE410T
21.	30W CHARGE	44.	WIFI_2.4G_5G & BT
22.	AOV	45.	GPS
23.	Test Points	46.	WCN3980



SDM630 GPIO Configuration

GPIO_0	BLSP_SPI_MOSI_1	GPIO_42	NC	GPIO_84	UIM2_CLK
GPIO_1	BLSP_SPI_MISO_1	GPIO_43	NC	GPIO_85	UIM2_RESET
GPIO_2	BLSP_SPI_CS1_N_1	GPIO_44	NC	GPIO_86	UIM2_PRESENT
GPIO_3	BLSP_SPI_CLK_1	GPIO_45	NC	GPIO_87	UIM1_DATA
GPIO_4	BLSP_UART_TX_2	GPIO_46	CAM0_RST_N	GPIO_88	UIM1_CLK
GPIO_5	BLSP_UART_RX_2	GPIO_47	CAM1_RST_N	GPIO_89	UIM1_RESET
GPIO_6	BLSP_I2C_SDA_2	GPIO_48	CAM2_RST_N	GPIO_90	UIM1_PRESENT
GPIO_7	BLSP_I2C_SCL_2	GPIO_49	NC	GPIO_91	UIM_BATT_ALARM
GPIO_8	BLSP_SPI_MOSI_3	GPIO_50	CAM_AF_VDD_EN	GPIO_92	NC
GPIO_9	BLSP_SPI_MISO_3	GPIO_51	CAM_AVDD_EN	GPIO_93	NC
GPIO_10	BLSP_SPI_CS_N_3	GPIO_52	NC	GPIO_94	NC
GPIO_11	BLSP_SPI_CLK_3	GPIO_53	LCD0 RESET_N	GPIO_95	NC
GPIO_12	NC	GPIO_54	SD_CARD_DET_N	GPIO_96	WDOG_DISABLE
GPIO_13	KEYPAD_LED_PWM	GPIO_55	DP_EN_N	GPIO_97	NC
GPIO_14	BLSP_I2C_SDA_4	GPIO_56	USBC_ORIENTATION	GPIO_98	NC
GPIO_15	BLSP_I2C_SCL_4	GPIO_57	FORCED_USB_BOOT	GPIO_99	QLINK_REQUEST
GPIO_16	BLSP_UART_TX_5	GPIO_58	USB_PHY_PS	GPIO_100	QLINK_ENABLE
GPIO_17	BLSP_UART_RX_5	GPIO_59	MDP_VSYNC_P	GPIO_101	RFFE1_DATA - SDR660 debug
GPIO_18	BLSP_UART_CTS_N_5	GPIO_60	NC	GPIO_102	RFFE1_CLK - SDR660 debug
GPIO_19	BLSP_UART_RFR_N_5	GPIO_61	NC	GPIO_103	RFFE2_DATA - (41)QAT3533
GPIO_20	FP_SUB_RESET	GPIO_62	NC	GPIO_104	RFFE2_CLK - (41)QAT3533
GPIO_21	SMB_STAT	GPIO_63	NC	GPIO_105	NC
GPIO_22	BLSP_I2C_SDA_6	GPIO_64	NC	GPIO_106	NC
GPIO_23	BLSP_I2C_SCL_6	GPIO_65	NC	GPIO_107	RFFE4_DATA-(45)QSW8574,(48)RF1656SR
GPIO_24	CDC_SWR_CLK	GPIO_66	TS_RESET_N	GPIO_108	RFFE4_CLK -(45)QSW8574,(48)RF1656SR
GPIO_25	CDC_SWR_DATA	GPIO_67	TS_INT_N	GPIO_109	RFFE5_DATA-(52)QAT3514,(52)QAT3550,(47) QAT3522
GPIO_26	WSA_SPKR_SD_N_1	GPIO_68	ACC_GYRO_DATA_AVA_INT_N	GPIO_110	RFFE5_CLK-(52)QAT3514,(52)QAT3550,(47)QAT3522
GPIO_27	NC	GPIO_69	ACC_GYRO_MOT_INT_N	GPIO_111	RFFE6_DATA -(46)QET4101 ch0,(41)QPA4361
GPIO_28	NFC_IRQ	GPIO_70	MAG_INT_N	GPIO_112	RFFE6_CLK -(46)QET4101 ch0,(41)QPA4361
GPIO_29	NFC_EN	GPIO_71	ALSP_INT_N	GPIO_113	NC
GPIO_30	NFC_DWL_REQ	GPIO_72	FP_INT_N_1	LPI_GPIO_0	LPI_SPI_1_CS1_N
GPIO_31	NFC_ESE_PWR_REQ	GPIO_73	NC	LPI_GPIO_1	LPI_PWR_EN
GPIO_32	CAM_MCLK0	GPIO_74	NC	LPI_GPIO_2	LPI_I2C_3_SDA
GPIO_33	CAM_MCLK1	GPIO_75	NC	LPI_GPIO_3	LPI_I2C_3_SCL
GPIO_34	CAM_MCLK2	GPIO_76	NC	LPI_GPIO_4	LPI_SPI_2_CS_N
GPIO_35	NC	GPIO_77	AUDIO_USBC_EN2_N	LPI_GPIO_5	LPI_SPI_2_CLK
GPIO_36	CCI_I2C_SDA0	GPIO_78	WMSS_RESETN	LPI_GPIO_6	LPI_SPI_2_MOSI
GPIO_37	CCI_I2C_SCL0	GPIO_79	PWM_CNTRL	LPI_GPIO_7	LPI_SPI_2_MISO
GPIO_38	CCI_I2C_SDA1	GPIO_80	AUDIO_USBC_EN1	LPI_GPIO_8	LPI_SPI_1_CS_N
GPIO_39	CCI_I2C_SCL1	GPIO_81	MSS_LTE_COXM_TXD	LPI_GPIO_9	LPI_SPI_1_CLK
GPIO_40	NC	GPIO_82	MSS_LTE_COXM_RXD	LPI_GPIO_10	LPI_SPI_1_MOSI
GPIO_41	FL_STROBE_TRIG	GPIO_83	UIM2_DATA	LPI_GPIO_11	LPI_SPI_1_MISO

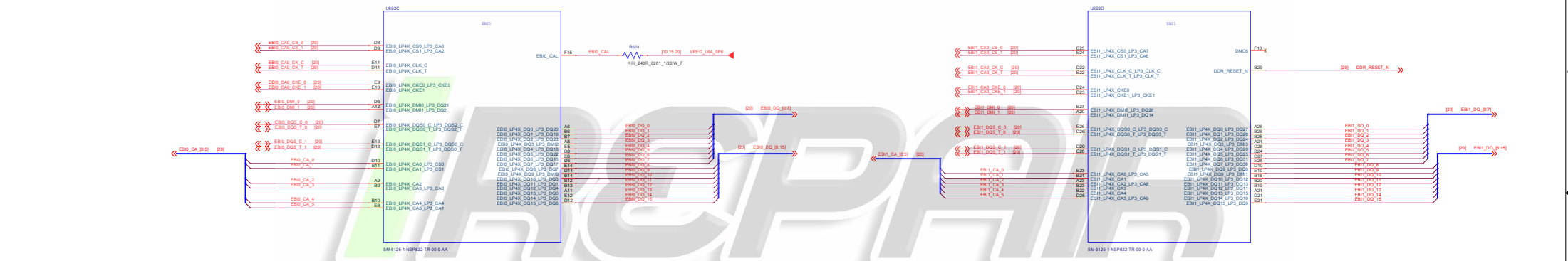
LPI_GPIO_12	LPI_UART_1_TX
LPI_GPIO_13	LPI_UART_1_RX
LPI_GPIO_14	NC
LPI_GPIO_15	NC
LPI_GPIO_16	NC
LPI_GPIO_17	NC
LPI_GPIO_18	CDC_PDM_CLK
LPI_GPIO_19	CDC_PDM_SYNC
LPI_GPIO_20	CDC_PDM_TX
LPI_GPIO_21	CDC_PDM_RX0
LPI_GPIO_22	CDC_PDM_RX0_COMP
LPI_GPIO_23	CDC_PDM_RX1
LPI_GPIO_24	CDC_PDM_RX1_COMP
LPI_GPIO_25	CDC_PDM_RX2
LPI_GPIO_26	CDC_DMIC_CLK1
LPI_GPIO_27	CDC_DMIC_DATA1
LPI_GPIO_28	CDC_DMIC_CLK2
LPI_GPIO_29	CDC_DMIC_DATA2
LPI_GPIO_30	LPI_QCA_SB_CLK
LPI_GPIO_31	LPI_QCA_SB_DATA0

PM660 GPIO Configuration

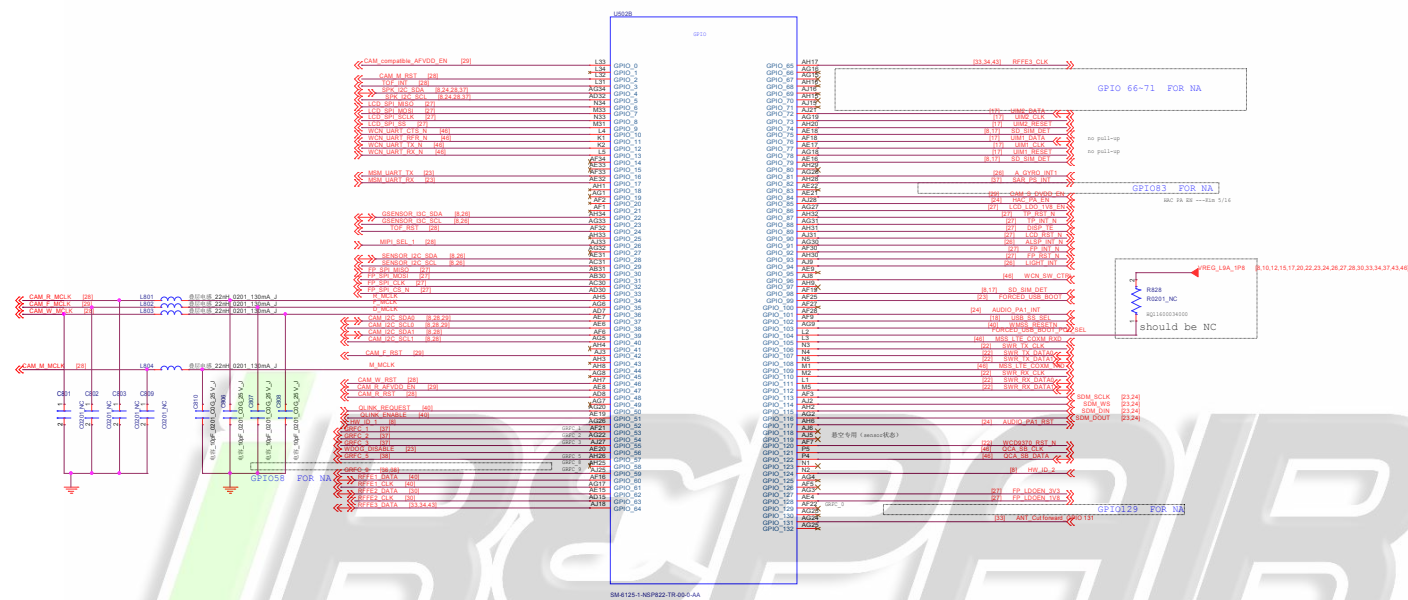
GPIO_1	OPTION1	GPIO_8	SLB
GPIO_2	DIV_CLK2	GPIO_9	uUSB_TYPEC
GPIO_3	NC	GPIO_10	WCSS_VCTRL
GPIO_4	NFC_CLK_REQ	GPIO_11	HOMEKEY_FP_PM_INT
GPIO_5	WLAN_SW_CTRL	GPIO_12	WIPWR_MODE
GPIO_6	SLP_CLK	GPIO_13	PM_A_GPIO_13
GPIO_7	UIM_BATT_ALARM		

PM660A GPIO Configuration

GPIO_1	OPTION2	GPIO_8	SD_CARD_DET_N
GPIO_2	LPI_PWR_EN	GPIO_9	WCSS_VCTRL
GPIO_3	FRONT_CAM_DVDD_EN	GPIO_10	SLB
GPIO_4	REAR_CAM_DVDD_EN	GPIO_11	LP4x_CTRL
GPIO_5	NC	GPIO_12	LP4x_MODE
GPIO_6	FP_VDD_EN		
GPIO_7	KEY_VOL_UP_N		



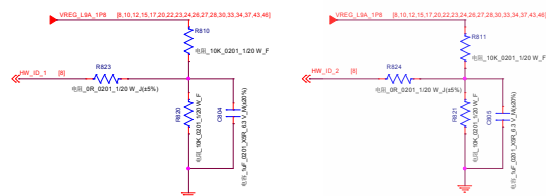
Company Name		
Stock Title		
Project		
Rev. 1.0		
Date: Sunday, October 24, 2021 8:00 AM		



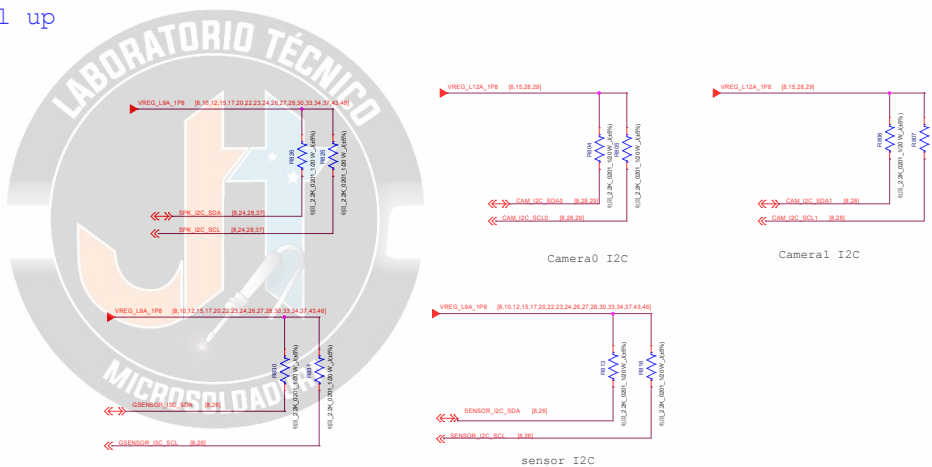
GPIO_56	BOOT_CONFIG(0)	WDOS_DISABLE
GPIO_87	BOOT_CONFIG(1)	FASTBOOT_SEL(0)
GPIO_32	BOOT_CONFIG(2)	FASTBOOT_SEL(1)
GPIO_118	BOOT_CONFIG(3)	FASTBOOT_SEL(2)
GPIO_99	FORCED_USB_BOOT	
GPIO_104	FORCED_USB_BOOT_POL_SEL	

BOOT OPTIONS	BOOT CONFIG BITS		
	3	2	1
0. Default (MMC->SD->USB/EDL)	0	0	0
1. SD->MMC->USB/EDL	0	0	1
2. SD->USB/EDL	0	1	0
3. USB/EDL	0	1	1
4. UPS->SD->USB/EDL	1	0	0
5. SD->UPS->USB/EDL	1	0	1
6. UPS->USB/EDL	1	1	0
7. UPS, PWM, MODE->USB/EDL	1	1	1

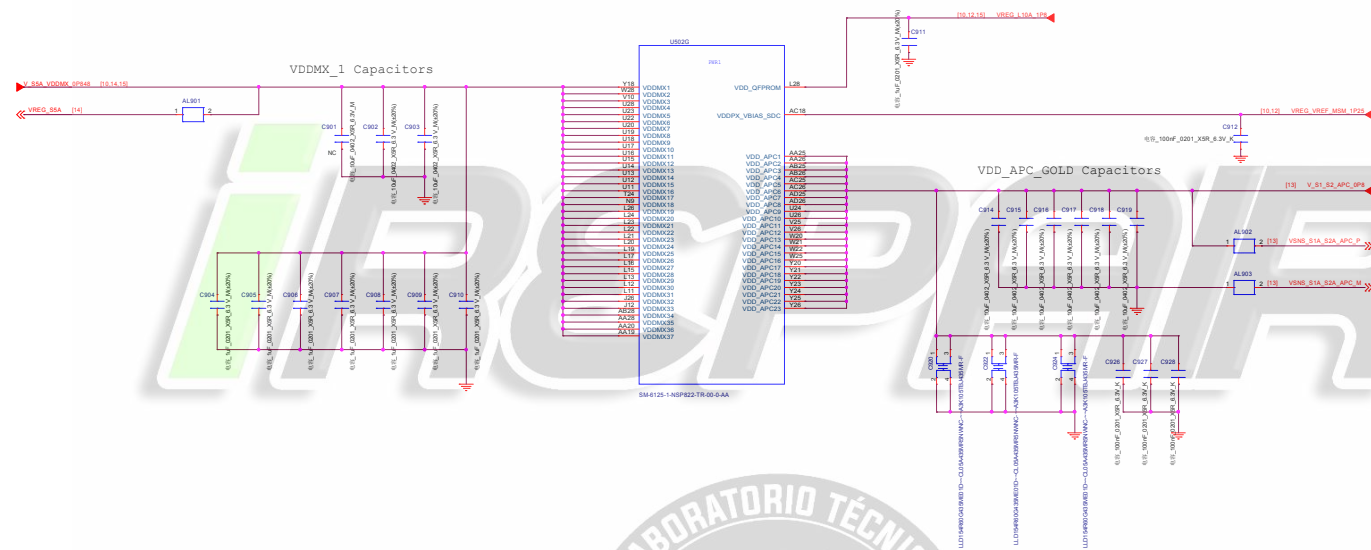
Hardware ID



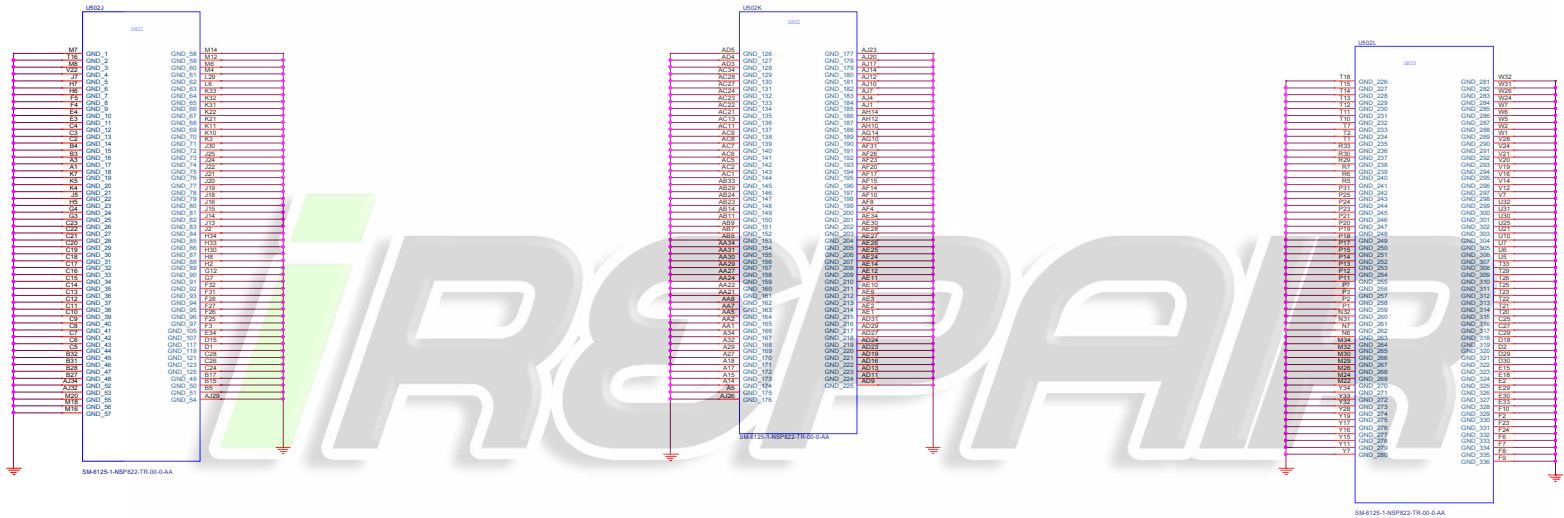
I2C pull up

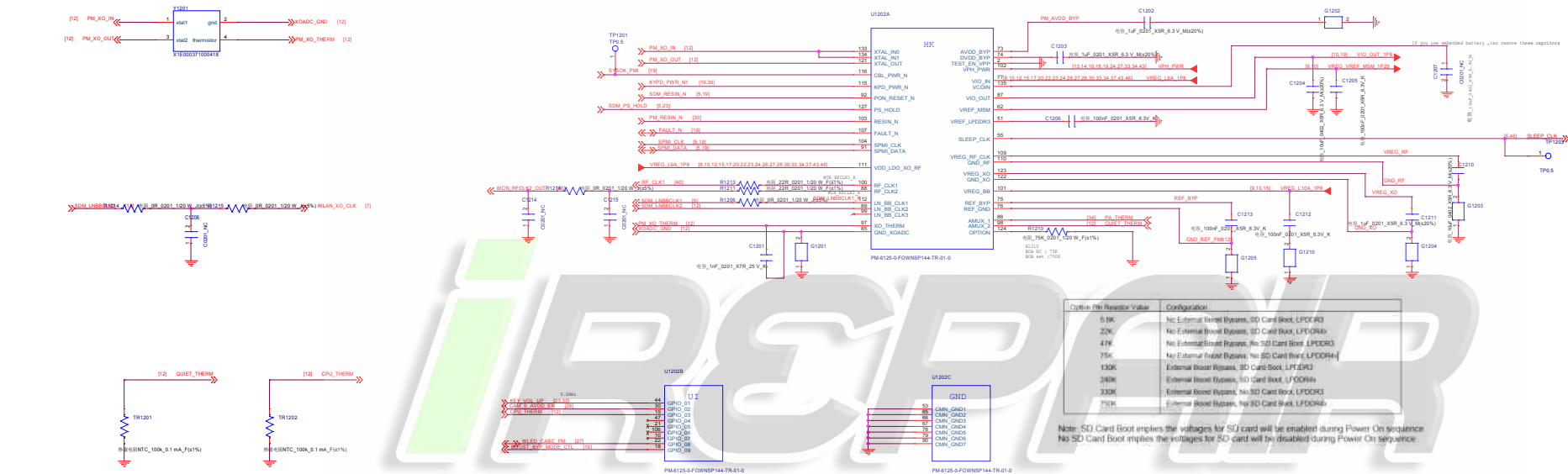


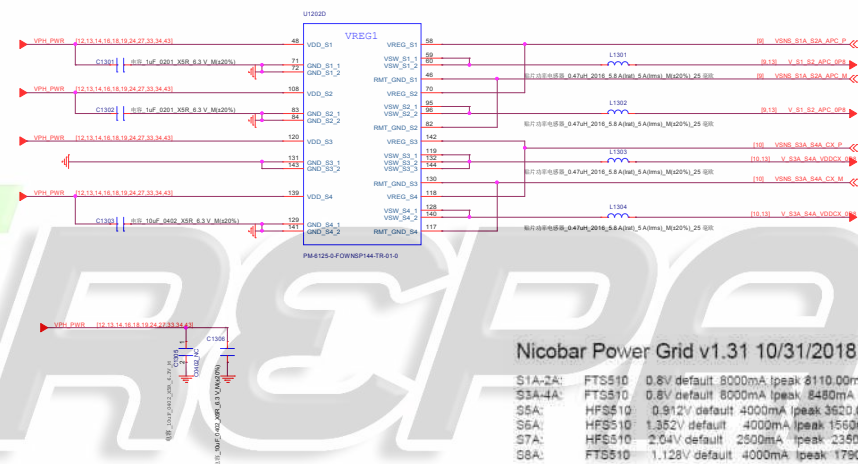
Company: Hsueh	
Block Title: SDMS30 GPIO (Internal Codec)	
Project: Teller	Rev: 1.0
Date: Sunday, December 06, 2009	Sheet: 9 of 48

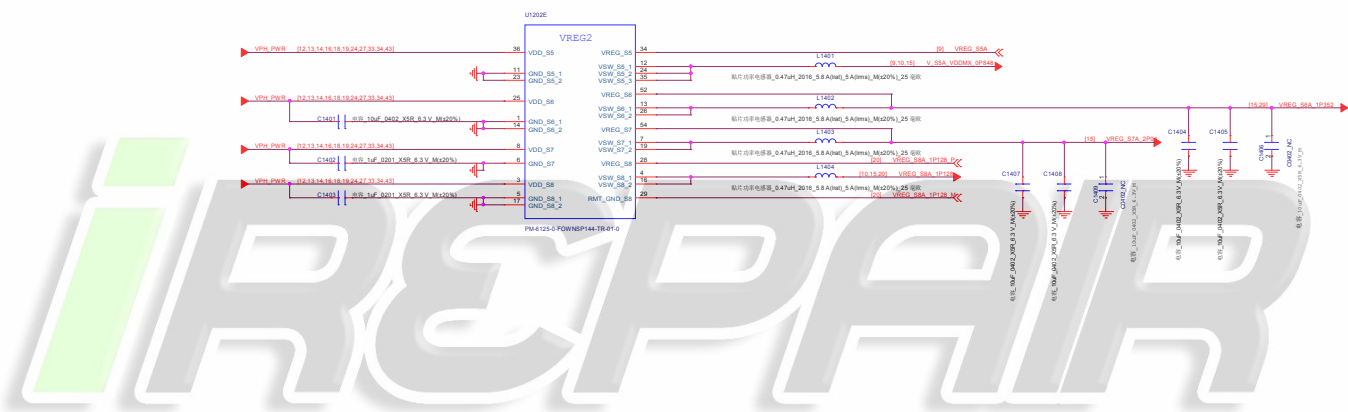


Company Hsien		
Block Title SM6125 Power (Core)		
Project	Taller	Rev 1.0
Date: Sunday, October 24, 2021 8:08:00 AM 9 of 16		

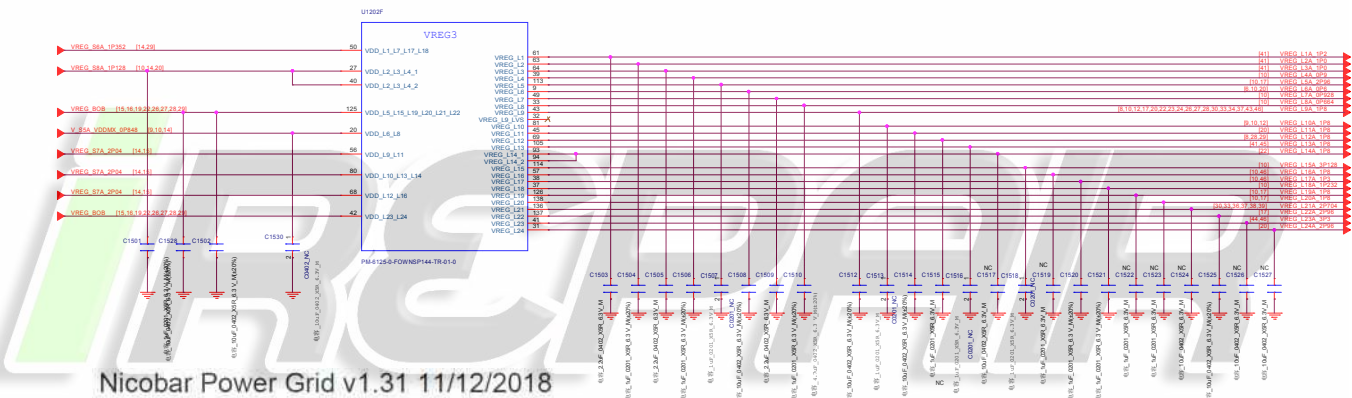








Company Hsien		
Stock Title	PM6125 LDO Circuits	
Project	Taller	Rev 1.0
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Nicobar Power Grid v1.31 11/12/2018

- LDO Type Source V_default (est) Target Pin#
- L1 N600 S8A 1.2V (625) QLN,SDR 1.2 61
 - L2 N300 S8A 1V (320) SDR VDD 1 0V 63
 - L3 N600 S8A 1V (370) SDR VDD 1 0V 64
 - L4 N300 S8A 0.928V (126) QLINK, EBI, MIFI 39
 - L5 MVP50 VPH 2.90V (22) PX 113
 - L6 N1300 S8A 0.924V (1040) VDDQ, EBI 9
 - L7 N600 S8A 0.928V (54-2) USB 49
 - L8 N300 S8A 0.664V (81) CX 33
 - L9 LVP600 STA 1.8V (754) Sensors, WCD, TMD, PMU, MSM, GET 43
 - L10 LVP150 STA 1.8V (100) QFPROM, USB, UFS, PX 81
 - L11 LVP600 STA 1.8V (925) EMMC, VDDQ 45
 - L12 LVP300 STA 1.8V (121) OV, USB, REDRIVER 69
 - L13 LVP300 STA 1.8V (204) QPM 1.8V, BBRX, GPS 105
 - L14 LVP600 STA 1.8V (650) WCD Buck 94, 93
 - L15 MVP150 VPH 3.128V (5.28) USB 3.1V, DP PHY 114
 - L16 LVP150 STA 1.8V (62.5) WCSS, WCN, XO 93
 - L17 N300 S8A 1.304V (246) WCN, WCSS, ADX DAC 39
 - L18 N300 S8A 1.202V (133) MIFI, VDDPX 37
 - L19 MVP150 VPH, BB 1.8V (60-2) Memory, SN100, PX 128
 - L20 MVP150 VPH, BB 1.8V (60-2) Memory, SN100, PX 138
 - L21 MVP600 VPH, BB 2.704V (0.68) QAT, DFE, RTC, QSW 136
 - L22 MVP600 VPH, BB 2.96V (800) MMC 137
 - L23 MVP600 VPH, BB 3.304V (591) WCN 3.3V 41
 - L24 MVP600 VPH, BB 2.96V (1170) EMMC 31

LDO L5A, L6L, L10L, L11V, L12L, L13L, L14L, L15L, L16L, L18L, L20L, L21V, L22A, L23L, L24 is the Pseudo-capless LDO, so can do CAP in BOM

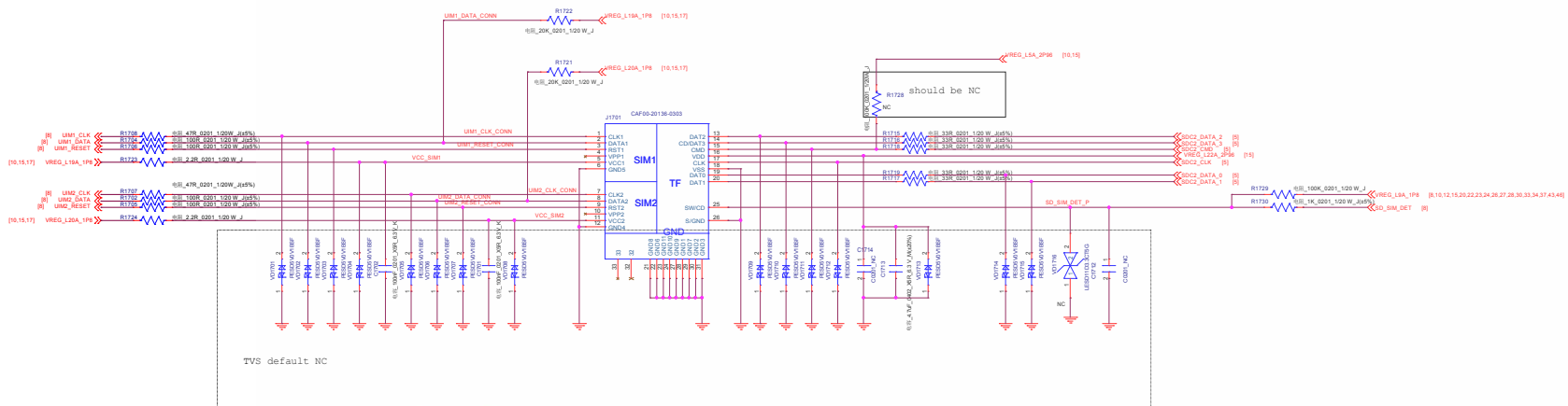
PSEUDO CAPLESS LDO:

P-type are pseudo-capless, cap can be at load

N-type CAPLESS LDO: If decoups on the load side do not add up by LDO spec, then install the cap close to the PMIC



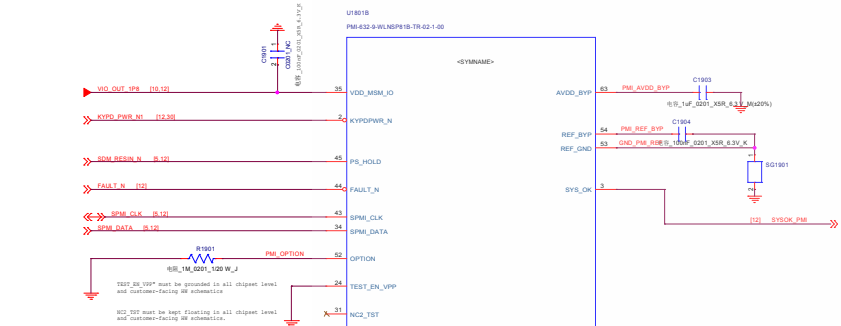
Company Name		
Stock Title	PM660 Charger_FG	
Project	Taller	Rev 1.0
Date	Sunday, October 24, 2016	Sheet 15 of 48



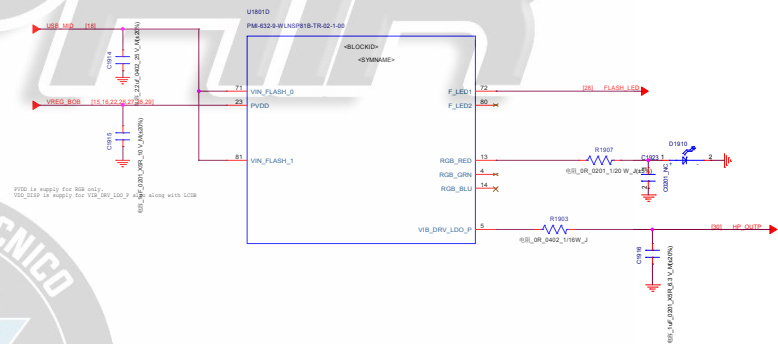
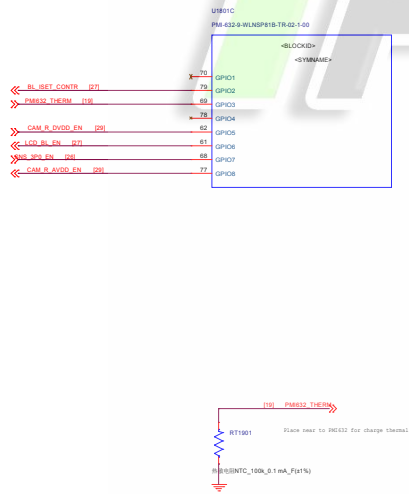
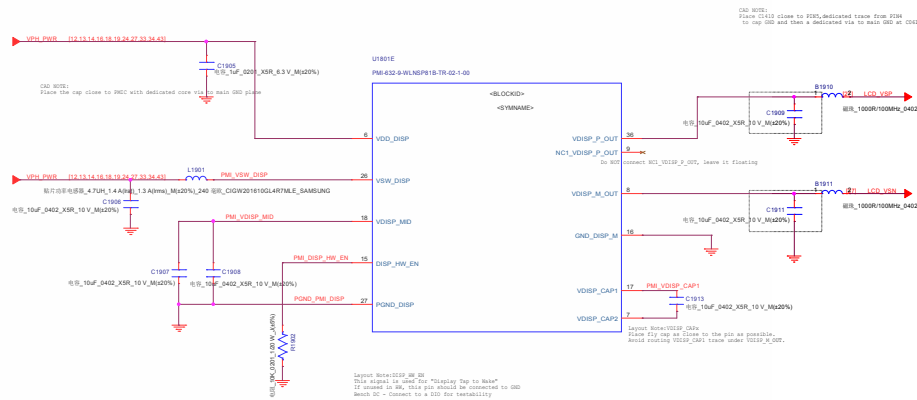
iREPAIR



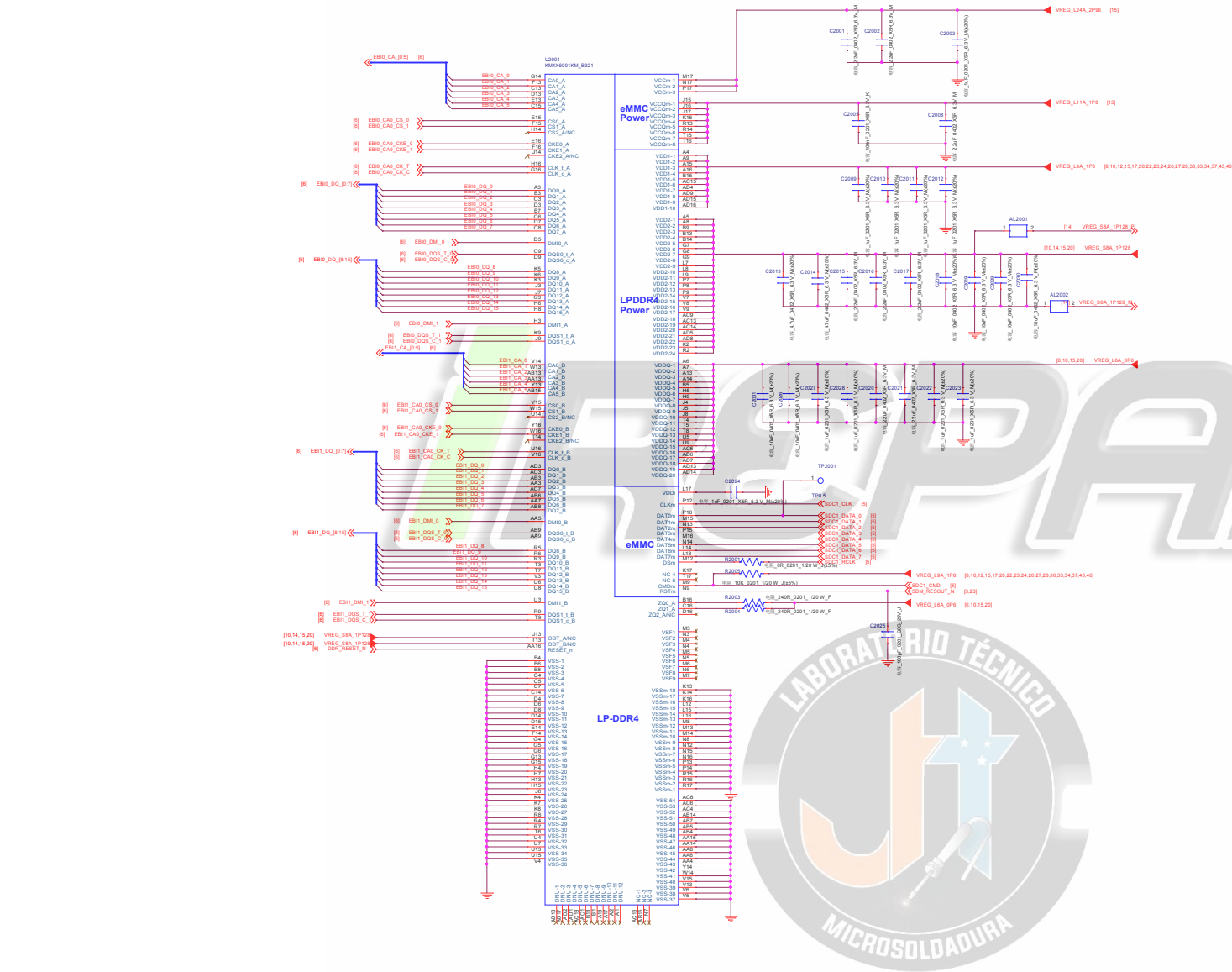
Company: Heagin		
Block Title: PM8501, Buck and LDO		
Project:	Taller	Rev: 1.0
Date:	Sunday, October 06, 2019	Sheet 17 of 46



OPTID	OPTION Configuration	IO Resistor
1	50K, 0508	1K
2	12K, 0508	22K
3	12K, TypeC	510K
	50K, TypeC	1K



Company: Huzjin		
Block Title: PM000L Display		
Project	Teller	Rev: 1.0
Date:	Sunday, October 08, 2012	Sheet: 12 of 48

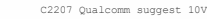


Company Header			
Stock Title		emCOP	
Project	Taller	Rev	1.0
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iREPAIR



Company Name			
Block Title 24W CHARGE			
Project	Taller	Rev	1.0
Date	Sunday, October 24, 2010	Sheet	21 of 46



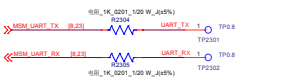
C2212/C2213/C2214/C2225 Qualcomm suggest resolution is K

Company: Hualin	
Block Title: Wipower	
Project: Teller	Rev: 1.0
Date: Sunday, October 06, 2019	Sheet: 22 of 48

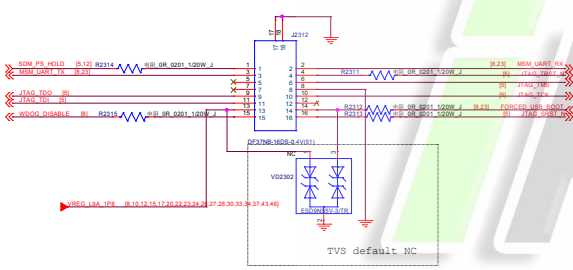
I2S test point



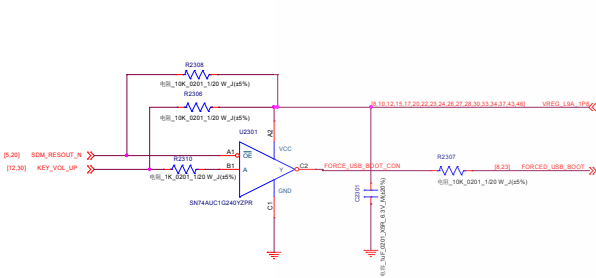
UART



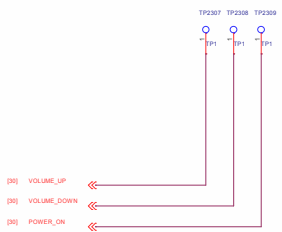
JTAG DEBUG CONNECTOR



power key +volup to enter 9008 mode



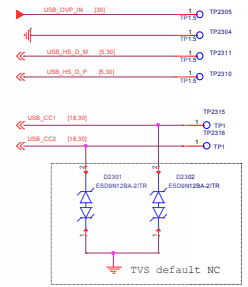
Side key



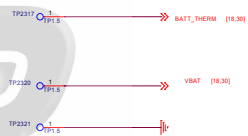
FORCE_BOOT



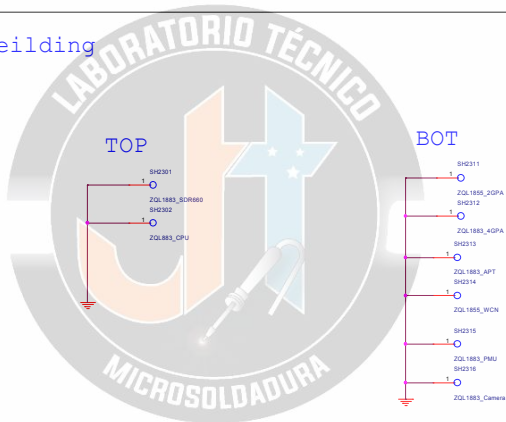
VBUS



VBAT



Shielding



MARK

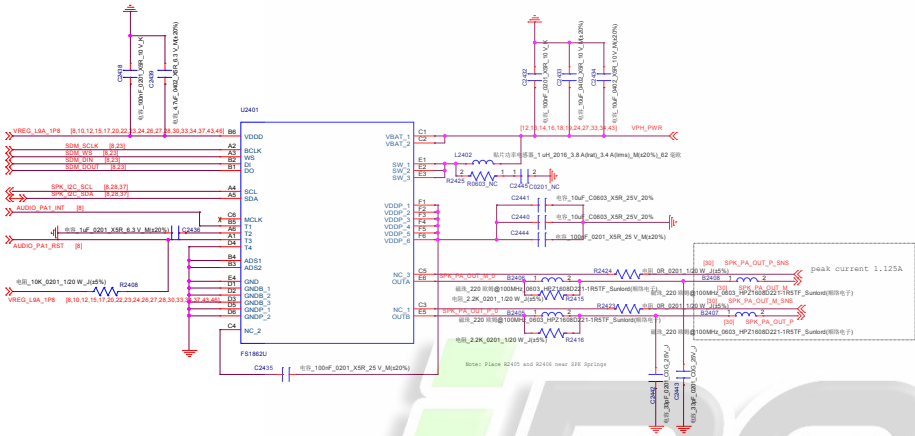


TP/GND/Shielding

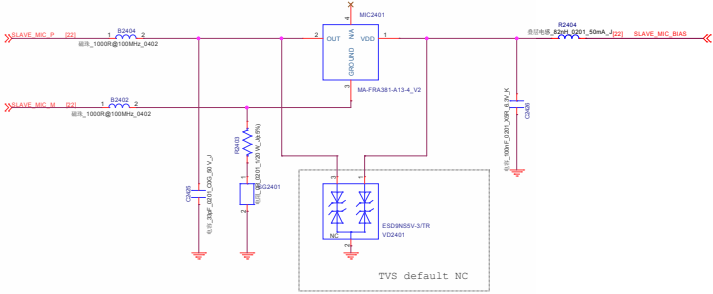
Company Hsien			
Stock Title	Test Points		
Project	Taller	Rev	1.0
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Speaker SMART PA

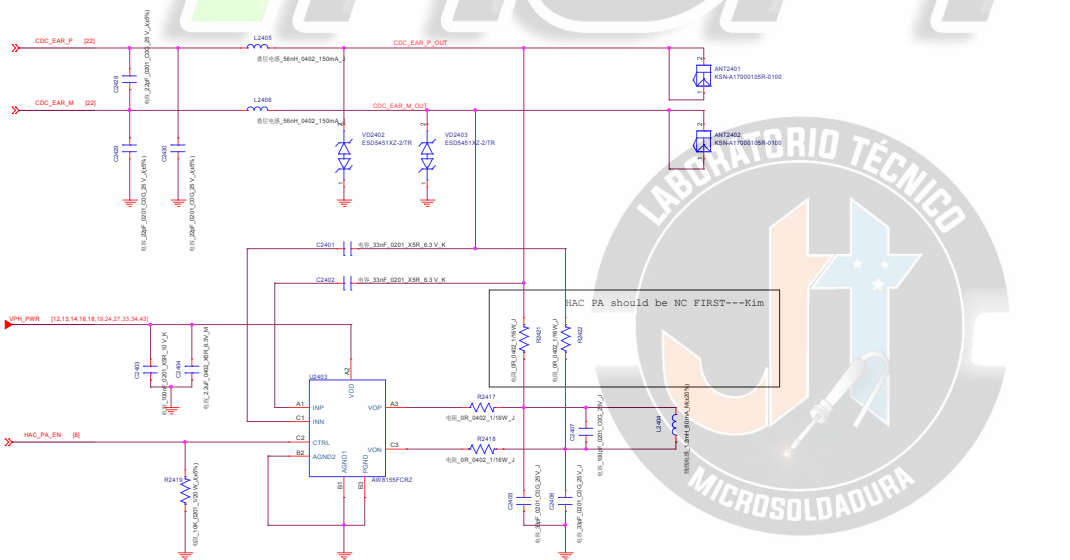
IIC Address
1nd TAS2558 0x4C
2ST FS1862 0x34



slave MIC



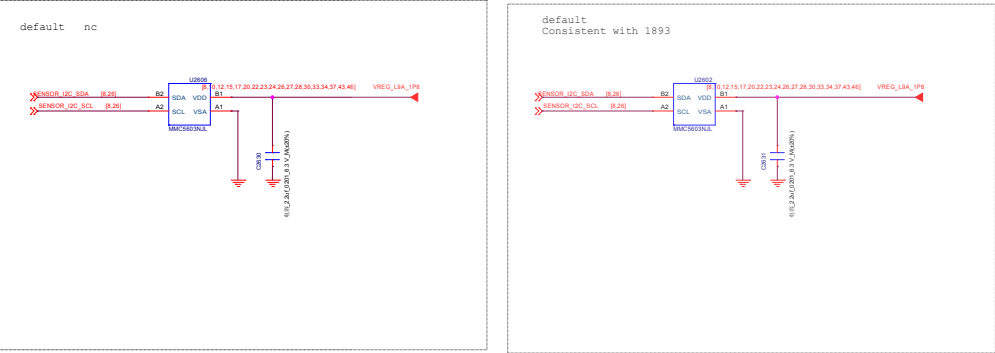
Receiver & HAC



HAC BOM list			
	HAC NC	HAC	
		PA NC	PA
C2401	NC	Y	Y
C2402	NC	Y	Y
C2403	NC	NC	Y
C2404	NC	NC	Y
C2405	NC	NC	Y
C2406	NC	Y	Y
C2407	NC	Y	Y
R2417	NC	NC	Y
R2418	NC	NC	Y
R2419	NC	NC	Y
R2421	NC	Y	NC
R2422	NC	Y	NC
L2404	NC	Y	Y
U2403	NC	N	Y

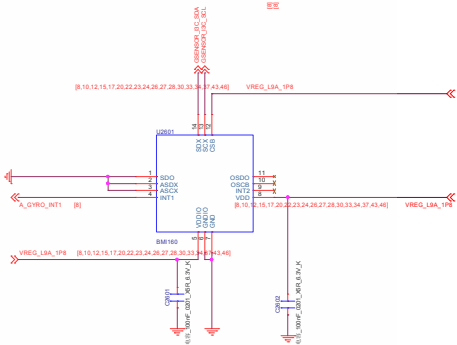
Magnetic Sensor

- 1st AK09918
- 2nd MMC5603NJL (Outline bigger)



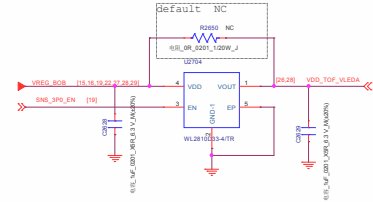
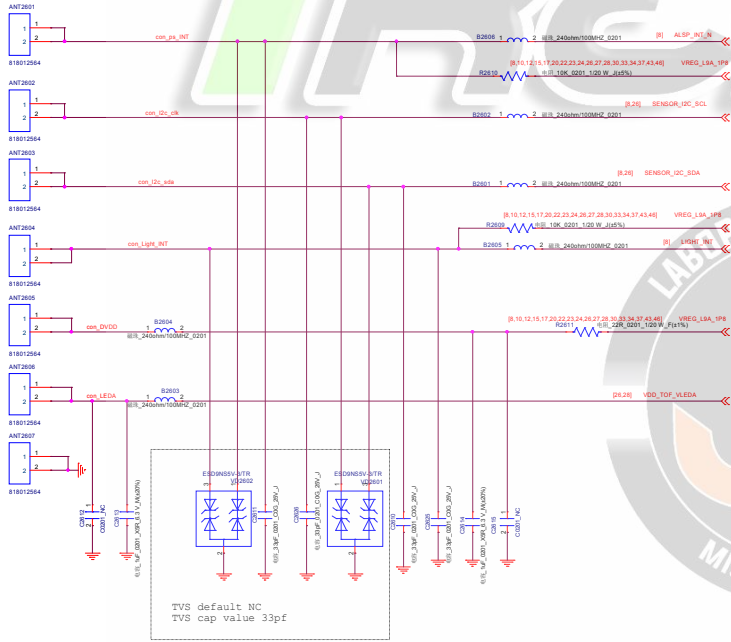
GRO Y Sensor

- 1st ST LSM6DSOW
- 2nd InvenSense ICM-42605



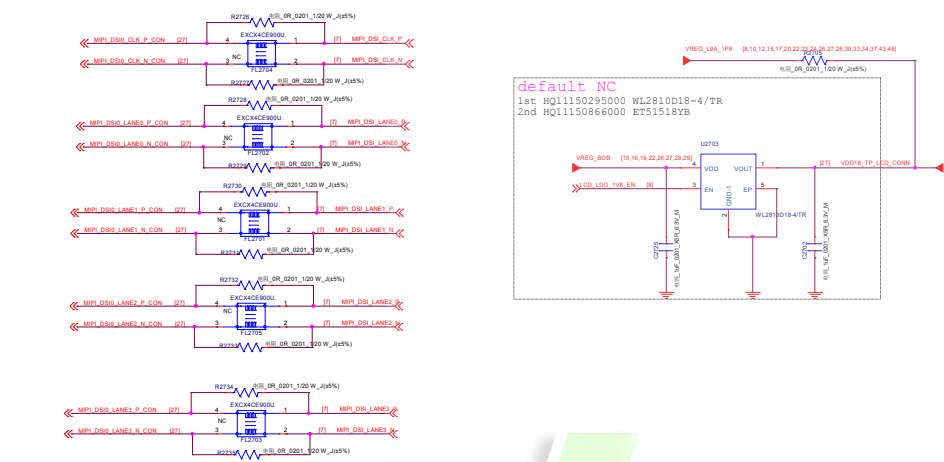
PS&ALS Sensor

- Psensor: 1st LTR-2568ALS-WA
- ALS: 1st TSL2540
- 2nd MN29005

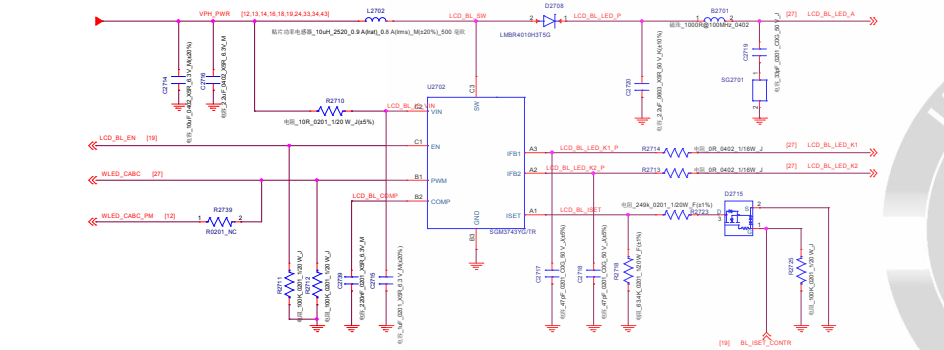


Company Hsien		
Stock Title	SENSORS MOTOR	
Project	Taller	Rev 1.0
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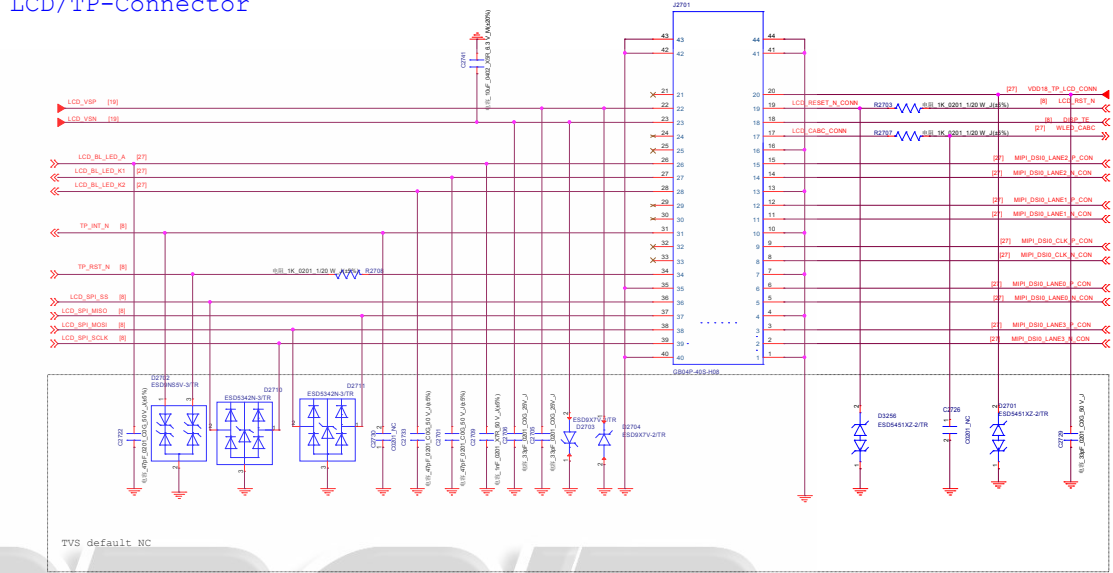
LCD MIPI Common Mode Filter
default EMIFilter



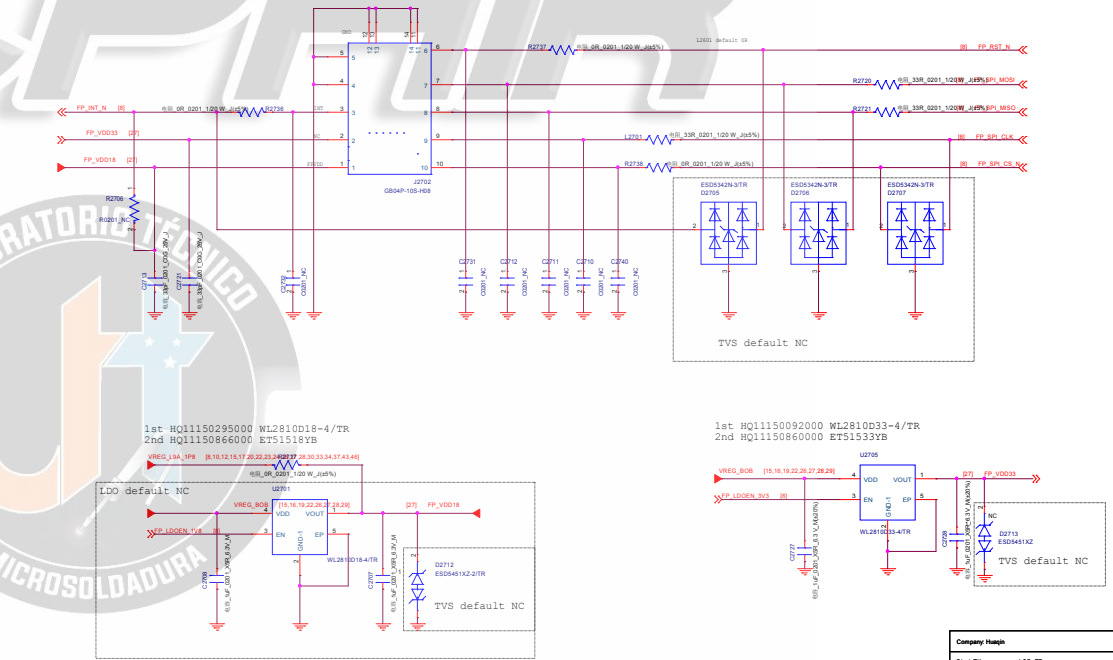
LCD BL driver
1st AW9963CSR HQ1151049000
2nd SGM3743YG/TR HQ11510166000



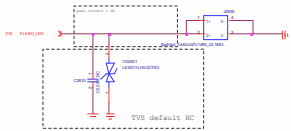
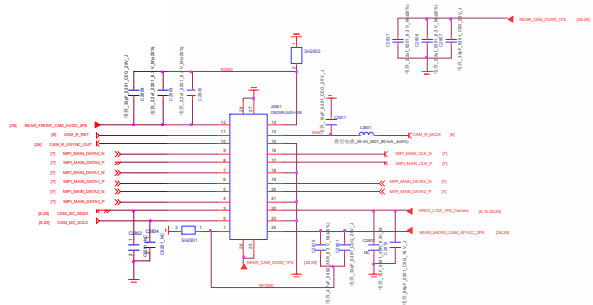
LCD/TP-Connector



FPS-Connector

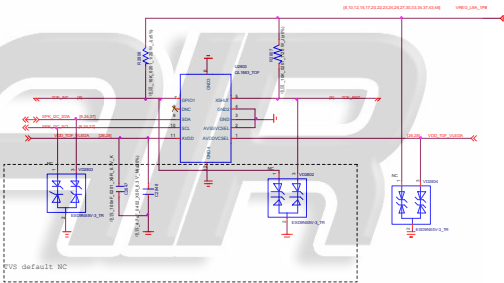
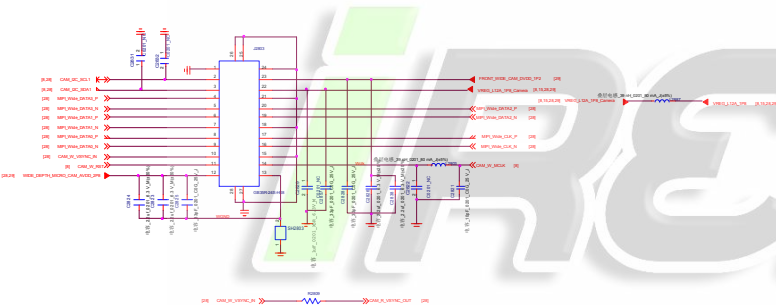


Company Header			
Block Title	LCD_TP		
Project	Taller	Rev	1.0
Date	Monday, October 24, 2022	Drawn	PT PT



WIDE CAMERA 8M

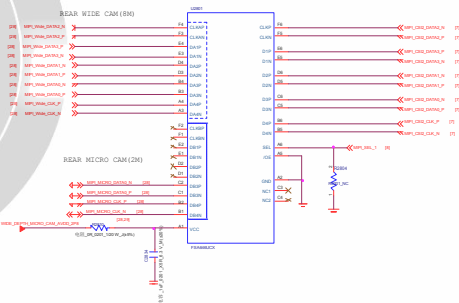
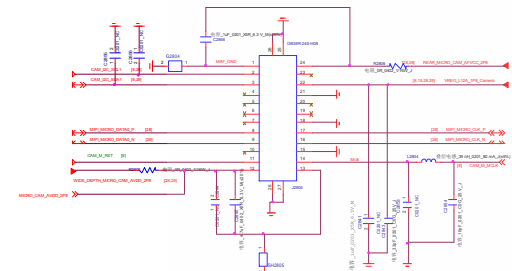
TOF



MICRO CAMERA 2M

8M&2M MIPI SWITCH

1st: RQ11140800000 9A84440-007K
2nd: RQ11170814000 9A84440-002X

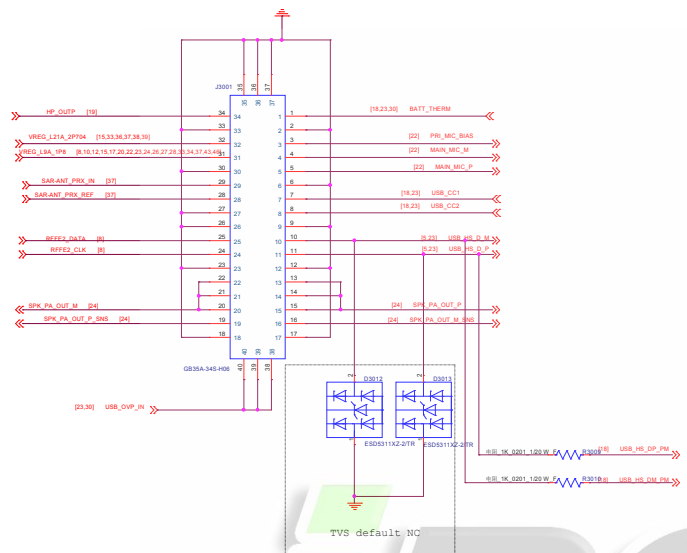


MIPI SWITCH		
	WIDE CAMERA (8M)	MICRO CAMERA(2M)
SEL	0	1

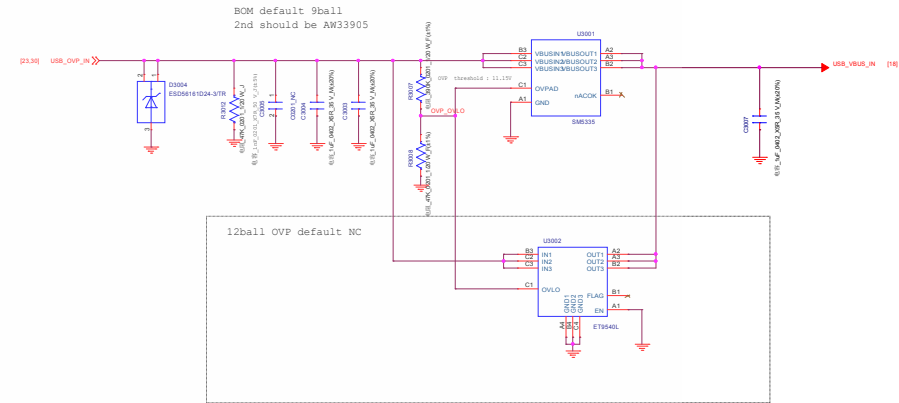
CSI lane mapping when a 4-lane interface is configured as 2-lane:
plus 1-lane interface:
* DCLK => DCLK_A
* DQ0 => DQ0_A
* DQ1 => DQ1_B
* DQ2 => DQ2_B
* DQ3 => DQ3_B

CSI lane mapping when a 4-lane interface is configured as 1-lane:
plus 1-lane interface:
* DCLK => DCLK_A
* DQ0 => DQ0_A
* DQ1 => DQ1_B
* DQ2 => DQ2_B
* DQ3 => DQ3_B

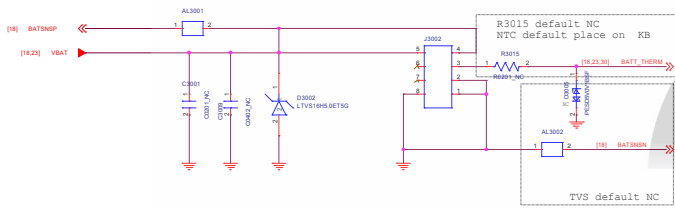
KEY_LINK_BATT



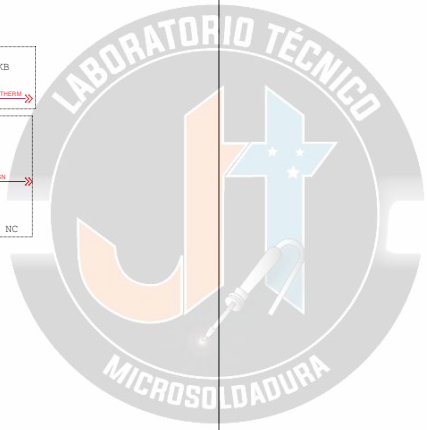
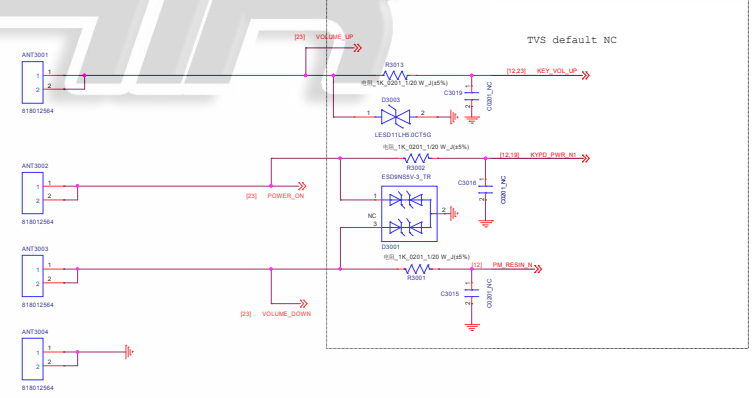
OVP



Battery Connector



Side KEY



Company Hsien			
Stock Title		SIM_SD_KEY_LINK_BATT	
Project		Taller	Rev 1.0
Date		Sunday, October 24, 2016	Sheet 50 of 66

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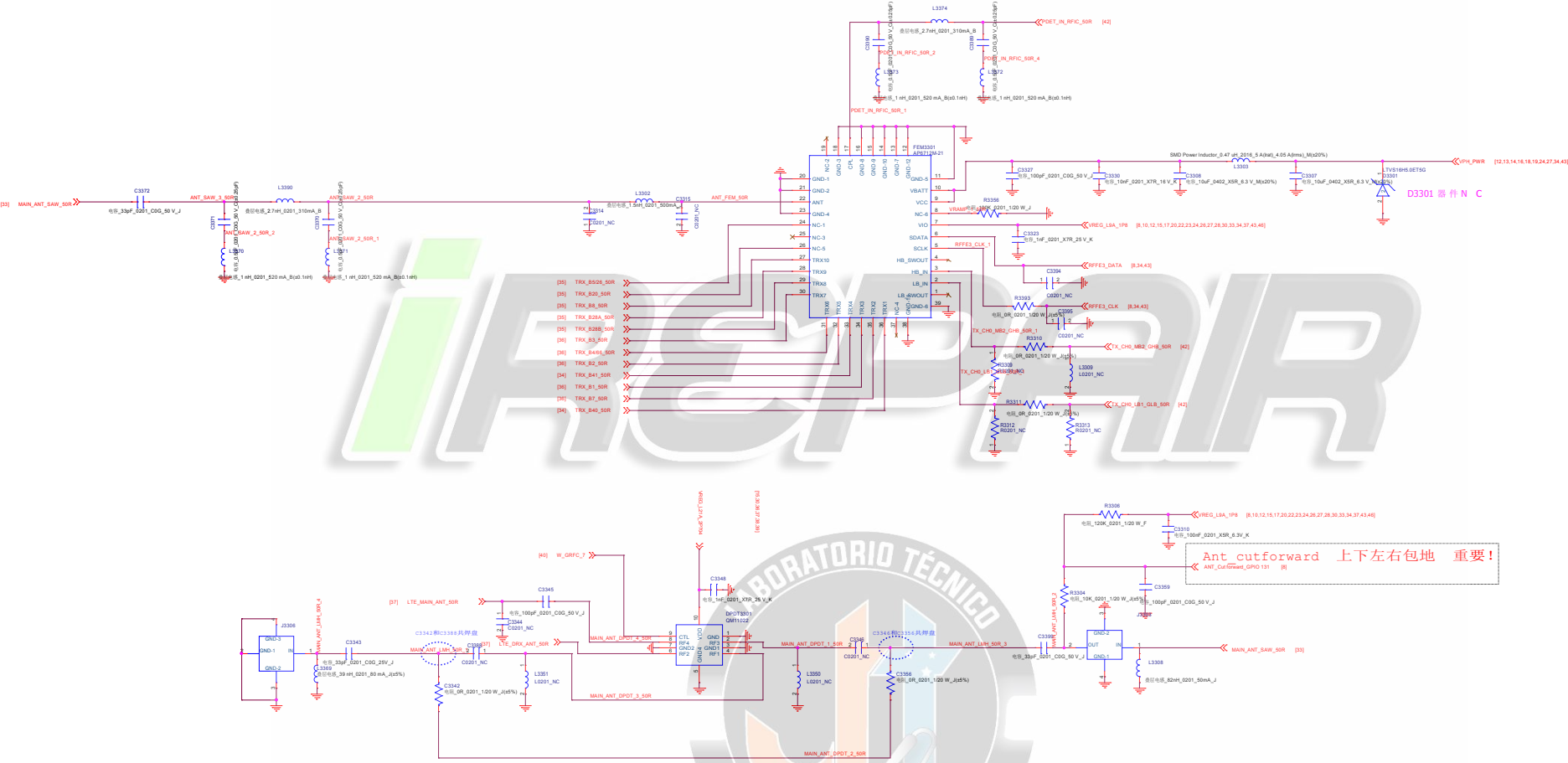
Company Name			
Block Title DTV			
Project	Taller	Rev	1.0
Date	Sunday, October 24, 2010	Sheet	51 of 66

PN553 I2C Adress:0x28 (Write:0x50 Read:0x51)



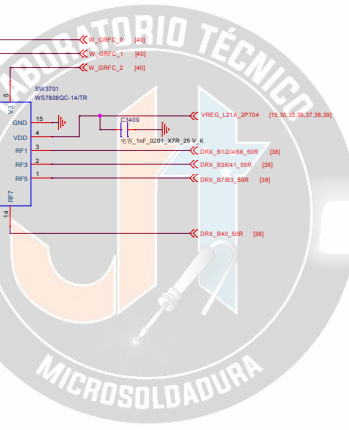
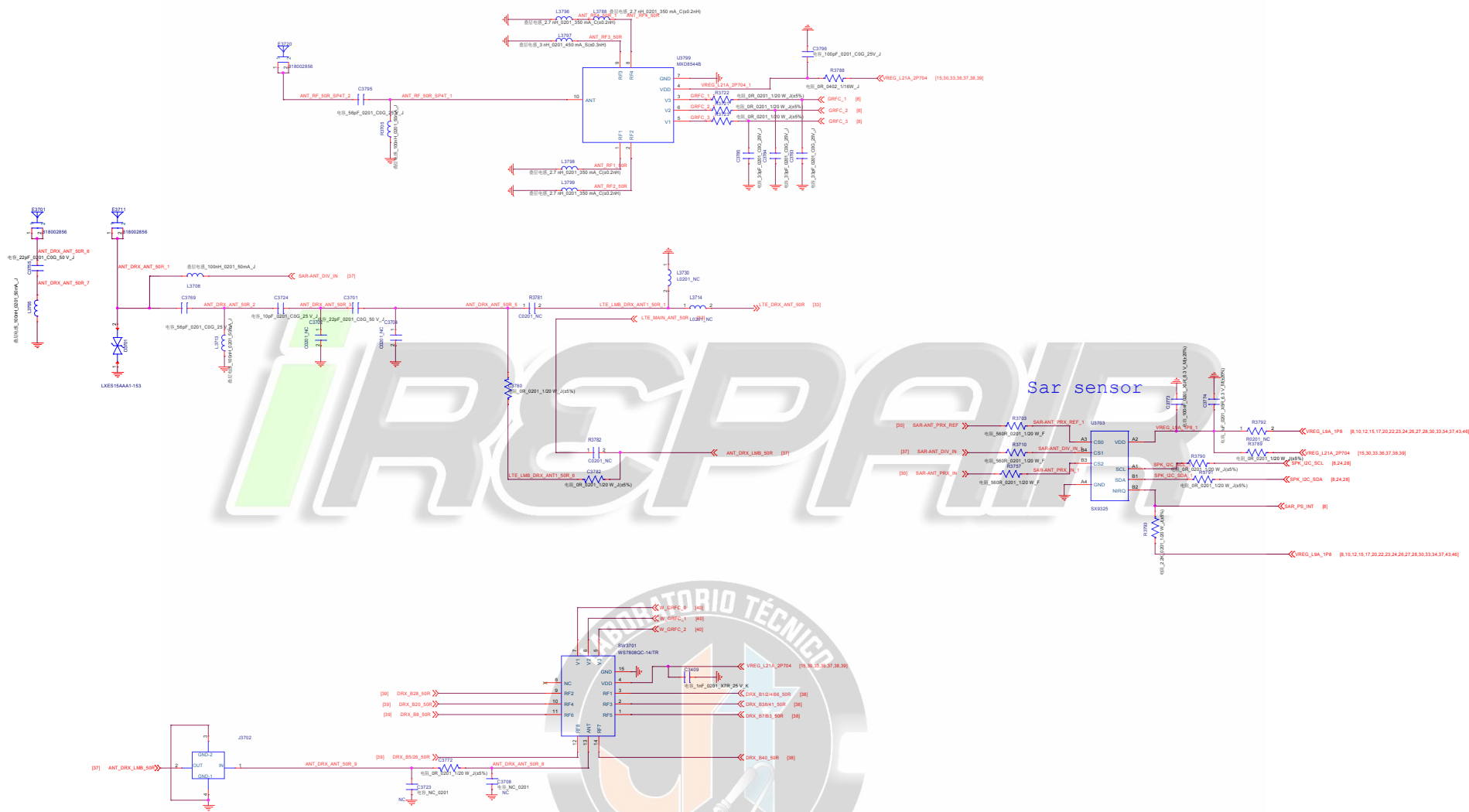
Company Name			
Stock Title		MFC	
Project	Taller	Rev	1.0
Date	Sunday, October 24, 2016	Sheet	50 of 66

PRIMARY_ANTENNA

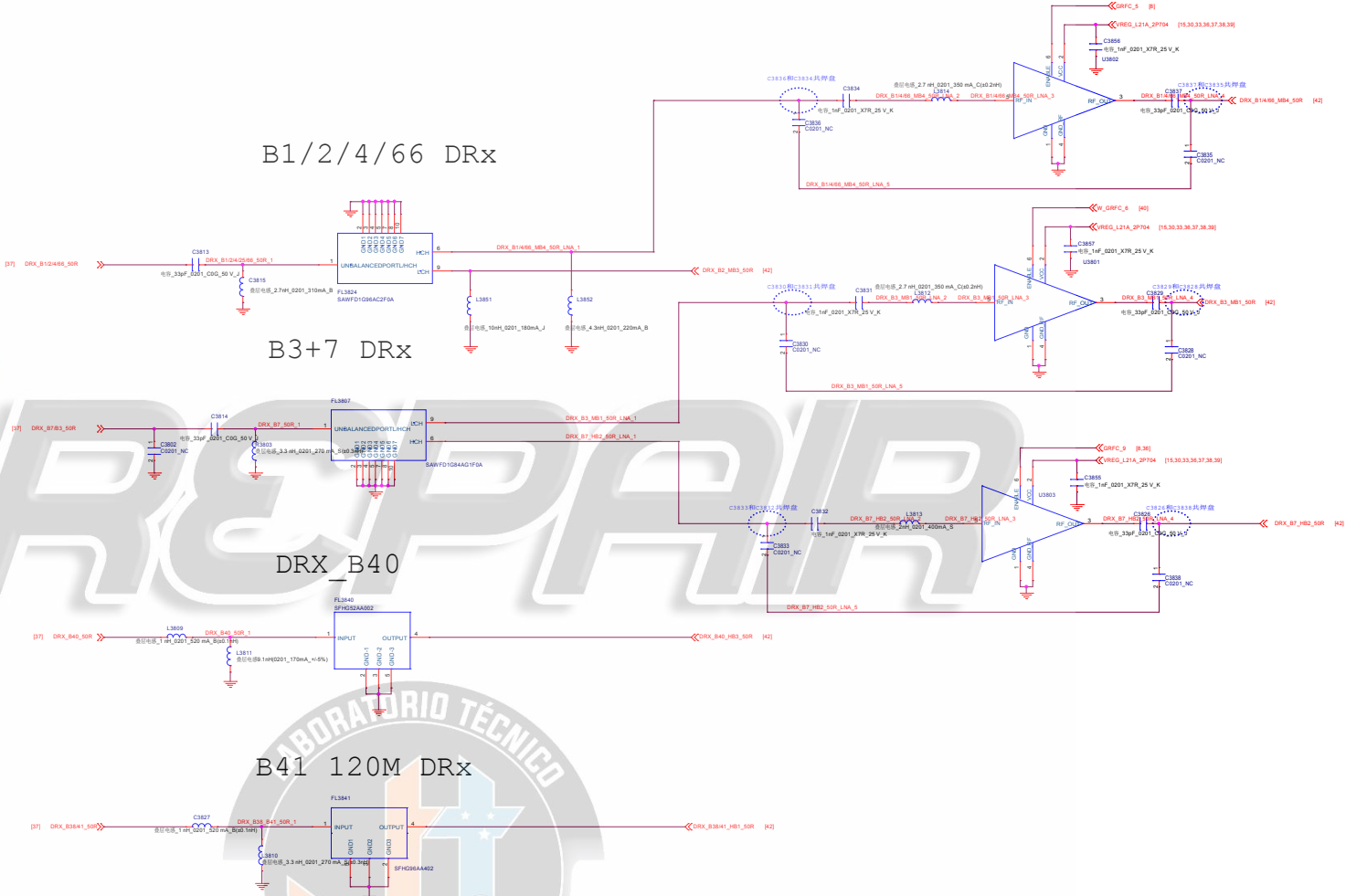


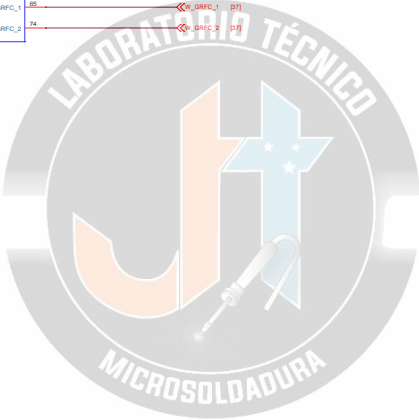
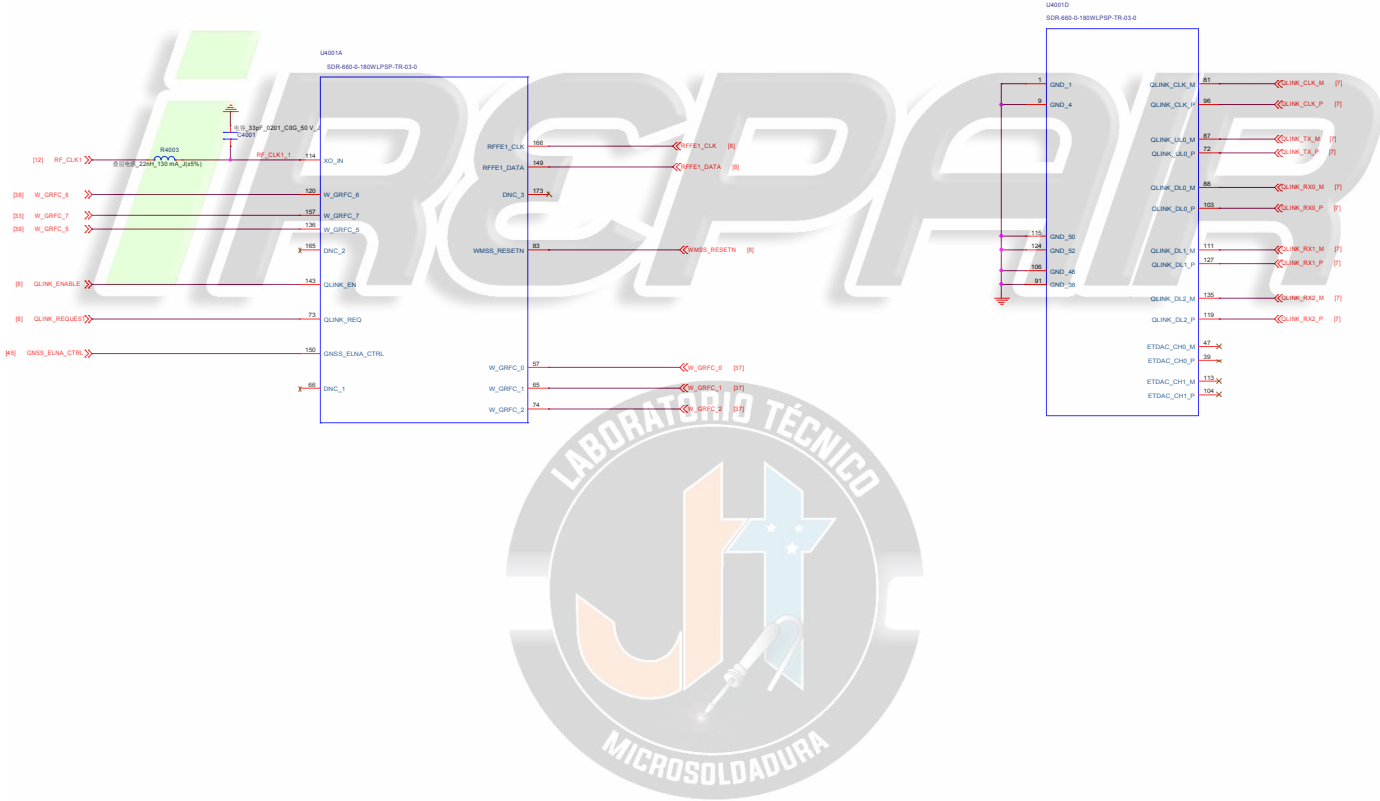
Ant cutforward 上下左右包地 重要!
ANT_CutForward_GPIO 131 [R]

Company Hsien			
Stock Title		PRIMARY_ANTENNA	
Project	Taller	Rev	1.0
Date	Sunday, October 24, 2010 8:00pm	Sheet	53 of 52

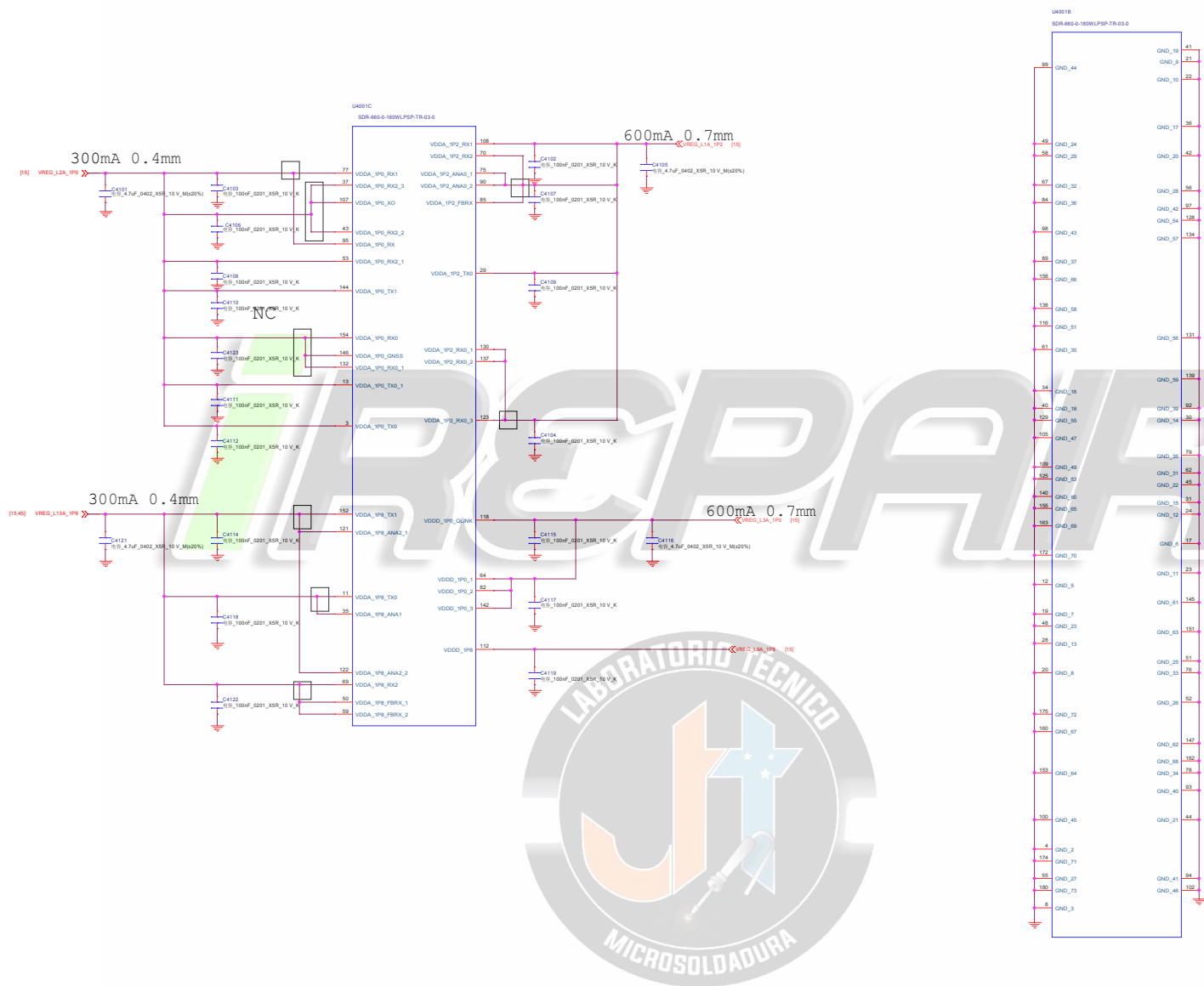


Company Hoop		
Block Title DIVERSIT_ANT		
Project	Taller	Rev 1.0
Date: Sunday, October 24, 2021 8:00 AM 10 of 30		



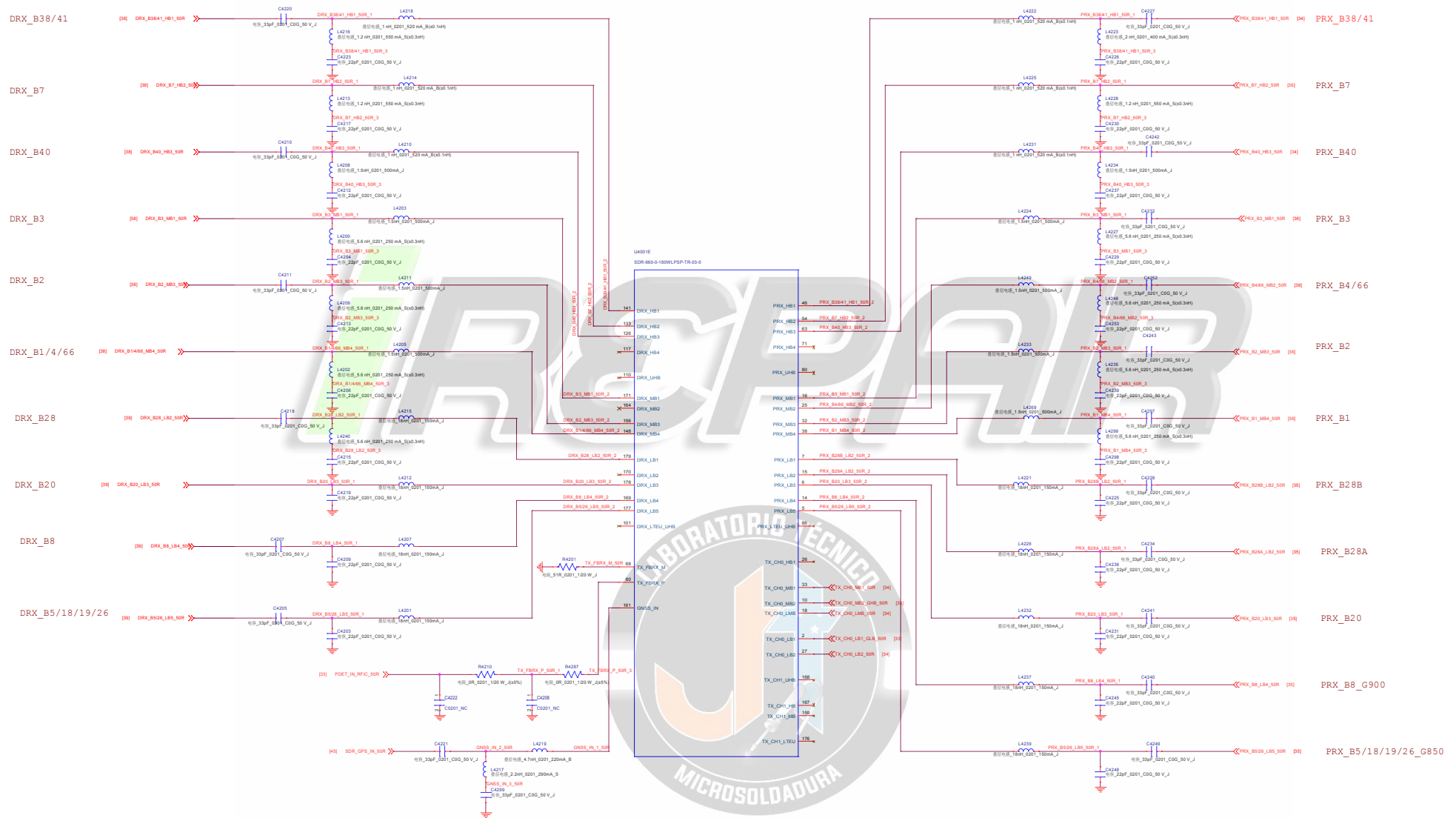


Company Hsien			
Block Title SDR660_MSS			
Project	Taller	Rev	1.0
Date	Sunday, October 24, 2022	Sheet	45 of 46

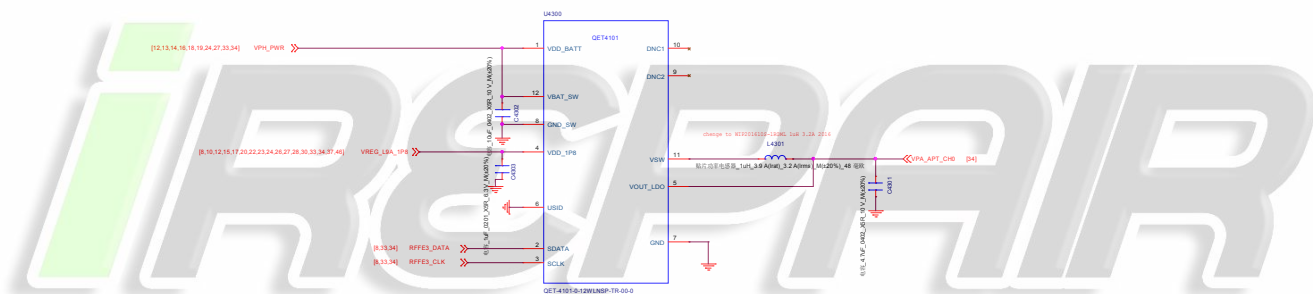


Company Name			
Block Title			
Project			
Date			

SDR660_DA/PRX/DRX/FBRX

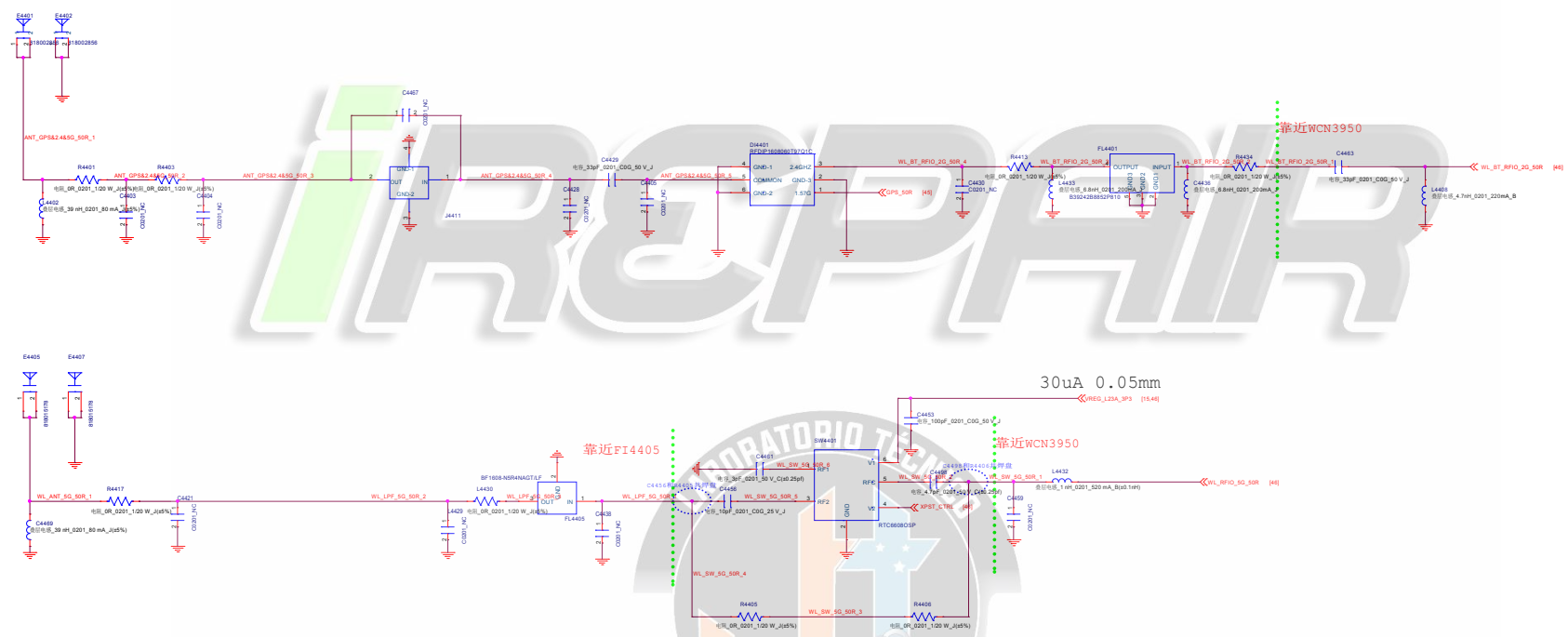
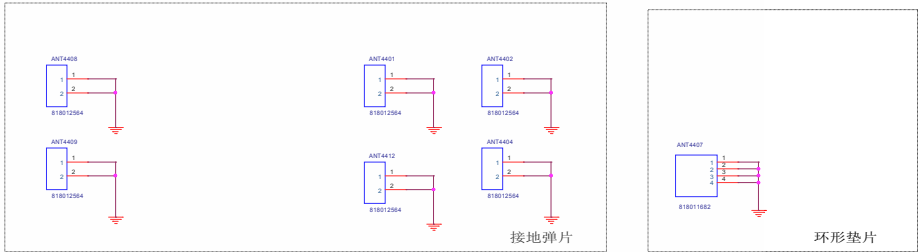


QET4101



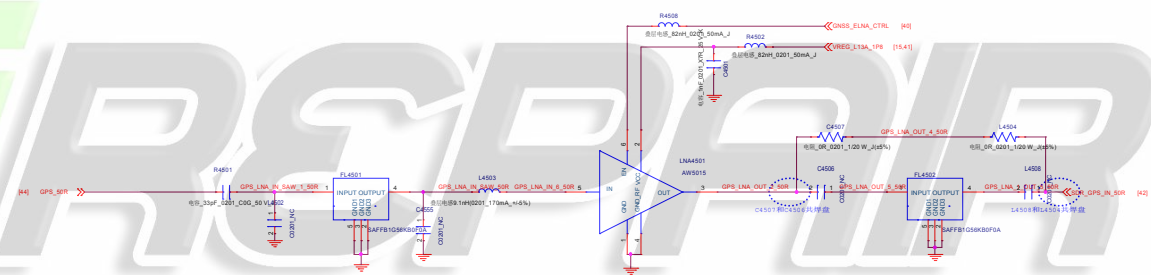
Company Hsien		
Stock Title QET4101		
Project	Taller	Rev 1.0
Date	Sunday, October 24, 2022	Sheet 43 of 48

WIFI 2.4G/BT



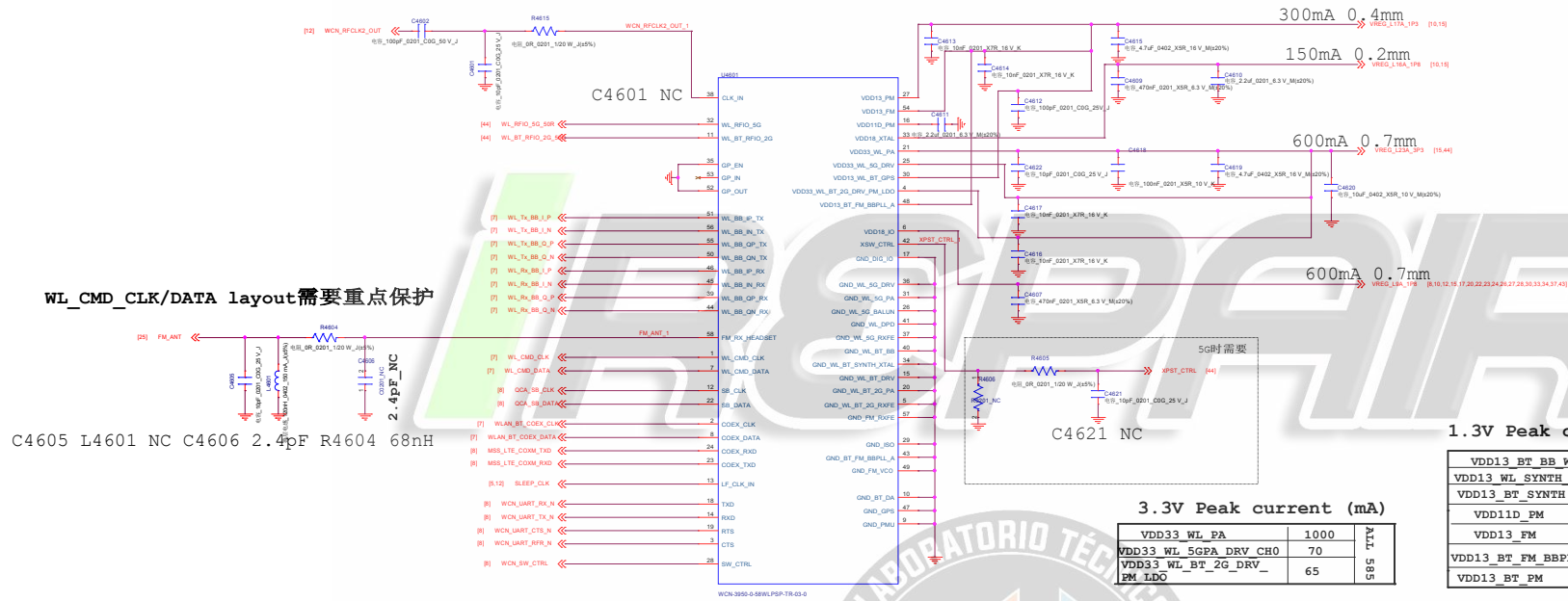
Company Hsien			
Block Title		WIFI 2.4G/BT	
Project	Taller	Rev	1.0
Date	Sunday, October 24, 2010	Sheet	43 of 48

GPS



WS7918DC Will

Company Hualin	
Block Title	GPS
Project	Teller
Rev	1.0
Date:	Sunday, October 08, 2012
Sheet	45 of 46



1.3V Peak current (mA)

VDD13_BT_BB_WL	120	ALL 240
VDD13_WL_SYNTH_CH0	40	
VDD13_BT_SYNTH	15	
VDD11D_FM	20	
VDD13_FM	15	ALL 75
VDD13_BT_FM_BBPLL	10	
VDD13_BT_FM	20	

1.8V Peak current (mA)

VDD18_IO	15	ALL 75
VDD18_XTAL	60	

3.3V Peak current (mA)

VDD33_WL_PA	1000	ALL 585
VDD33_WL_SGPA_DRV_CH0	70	
VDD33_WL_BT_2G_DRV_FM_LDO	65	

iREPAIR



Company Name			
Block Title DTV			
Project	Taller	Rev	1.0
Date	Sunday, October 24, 2010	Sheet	43 of 48